

Trip Analysis — Findings & Recommendations

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Overview

Statistical analysis of 200 rides tested whether fare pricing, surge multipliers, and timing relate to ride category, customer satisfaction, or revenue. Methods included independent-samples t-tests, Spearman correlation, IQR/z-score outlier detection, conditional probability, Bayes' theorem, and bootstrap confidence intervals.

Key Findings

Question Tested	Result	Interpretation
Premium vs Economy fare	t = -0.41, p = 0.684	No significant fare difference by category
Surge multiplier vs rating	ρ = -0.047, p = 0.508	No relationship between surge and satisfaction
Peak vs Non-Peak fare	t = -0.60, p = 0.550	Time of day does not drive fare amount
Fare anomalies (IQR / z-score)	1 clear outlier	₹37.59 fare on a 12-unit Economy trip — likely pricing error
Category demand share	Economy 41%, Shared 30%, Premium 29%	Economy dominates overall and peak-hour demand

Recommendations

- Investigate the ₹37.59 Economy fare (row 10, 12-unit trip, no surge) as a likely pricing or logging error before it feeds into downstream reporting.
- Current category-based pricing (Economy/Premium/Shared) shows no statistically significant fare differentiation — review whether the pricing model is achieving its intended segmentation.
- Surge pricing shows no measurable link to customer ratings (positive or negative) — surge can likely be adjusted for revenue purposes without a detectable satisfaction penalty, though this should be validated on a larger sample.
- Since Economy holds the largest demand share (41%, and highest even during peak hours), prioritize service reliability and driver availability for this segment to protect overall satisfaction.
- Sample sizes per group (n = 58–103) limit statistical power; treat 'no significant difference' findings as inconclusive rather than confirmed equivalence, and re-test as more data accumulates.

Caveat

All p-values exceeded 0.05, so no result here should be read as proof of 'no effect' — only that this sample did not detect one. Bootstrap resampling (10,000 iterations) confirmed the parametric test results, adding confidence in the reported estimates given the sample size available.